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Report On

Virtual Lab

BEC456B

Submitted By

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IV Semester
C Section

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

EXPERIMENT NO. 03 – LAB REPORT

CHARACTERIZATION AND EXTRACTION OF VARIOUS PARAMETERS OF µA741 IC

Subject	LIC (BEC456B)	Semester	IV
Experiment No.	03	Date	Feb 2026

1. AIM

To characterize and extract the Op-Amp IC µA741 parameters using Virtual Lab simulation and compare the measured values with the datasheet specifications.

2. APPARATUS USED

IC µA741 | Multimeter (DMM) | Power Supply (±15V) | Resistors (Rf = 470 KΩ, R1 = 470 KΩ, Rf = 10 KΩ, R1 = R2 = 100 Ω) | Capacitors (C1 = C2 = 1000 pF) | CRO | A.F. Oscillator | Connecting Cords

3. THEORY

3.1 Input Bias Current (I

The inverting (pin 2) and non-inverting (pin 3) terminals of the µA741 op-amp are the base terminals of the internal differential amplifier transistors. Ideally, no current should flow into these terminals. However, in practice, small base currents (**IB1** and **IB2**) flow at these terminals. The **Input Bias Current** is defined as the arithmetic average of these two currents:

$$IB = (IB1 + IB2) / 2$$

The bias current **IB1** at the inverting terminal is measured by noting the output voltage **Vo** across the feedback resistor **Rf = 470 KΩ**, and using Ohm's law: **IB1 = Vo / Rf**

The bias current **IB2** at the non-inverting terminal is measured similarly using **R1 = 470 KΩ**: **IB2 = Vo / R1**

3.2 Input Offset Current (I

Since the two bias currents **IB1** and **IB2** are not perfectly equal due to transistor mismatches, the **Input Offset Current** is defined as the absolute difference between them:

$$IOS = |IB1 - IB2|$$

Typical datasheet value: 20 nA; Maximum: 200 nA.

3.3 Input Offset Voltage (V

Even when both input terminals are connected to ground (0V), the output of a real op-amp may not be zero. This is due to internal circuit imbalances. **Input Offset Voltage** is the small differential voltage that must be applied between the inputs to force the output to zero. It is calculated from the closed-loop output using:

$$Vio = (Vo - Ios \times Rf) / (1 + Rf/R1)$$

Typical datasheet value: 2 mV; Maximum: 6 mV.

3.4 Output Offset Voltage (V

The **Output Offset Voltage** is the voltage appearing at the output (pin 6) when both inputs are grounded. It equals the input offset voltage multiplied by the closed-loop gain of the circuit:

$$VOOS = VIOS \times (1 + Rf/R1)$$

This was directly observed on the DMM as the output of the offset voltage measurement circuit.

3.5 Slew Rate (SR)

The **Slew Rate** is defined as the maximum rate of change of the output voltage in response to a step or large-signal input. It is expressed in V/µs:

$$SR = 2\pi \times fmax \times Vm / 10^6 \quad V/\mu s$$

where **fmax** is the maximum frequency (in KHz) at which the output remains undistorted (before becoming triangular), and **Vm = 3V** is the input signal amplitude. The slew rate of µA741C from the datasheet is typically **0.5 V/µs**.

4. CIRCUIT DESCRIPTIONS

Inverting bias circuit: µA741 with Rf = 470 KΩ in feedback, C1 = 1000 pF, supply ±15V. DMM connected across output to measure Vo.

Non-inverting bias circuit: µA741 with R1 = 470 KΩ at non-inverting input, C1 = 1000 pF, supply ±15V.

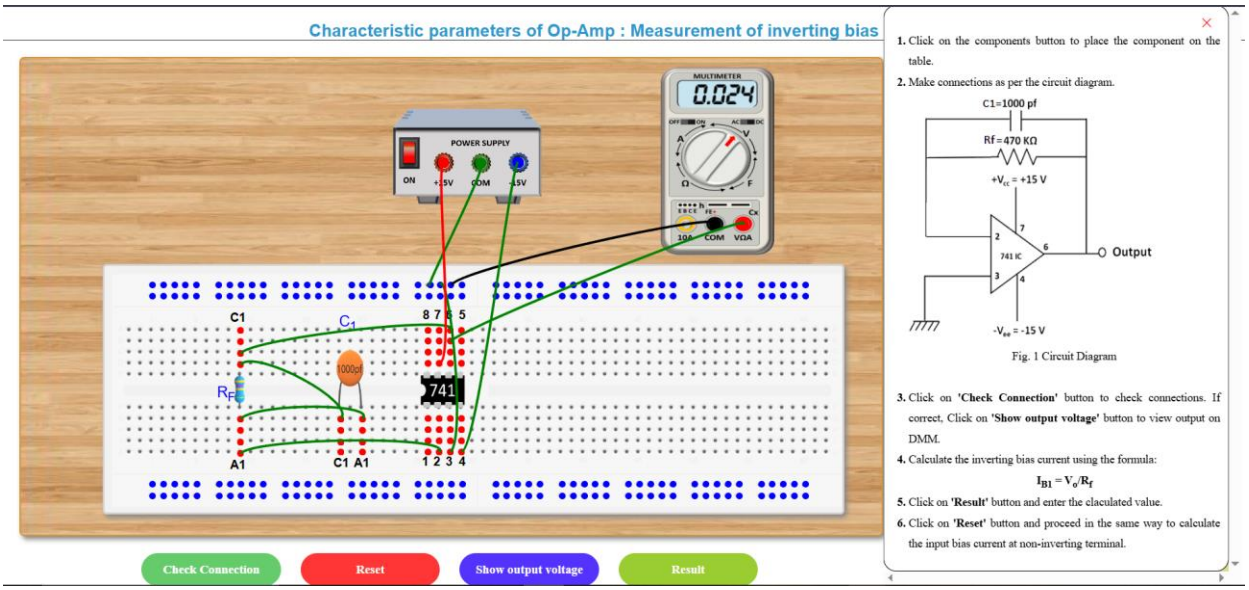
Input offset current circuit: Both feedback resistors Rf = 470 KΩ and R1 = 470 KΩ connected with capacitors C1 = C2 = 1000 pF.

Input offset voltage circuit: µA741 with Rf = 10 KΩ, R1 = R2 = 100 Ω, C1 = 1000 pF, supply ±15V.

Slew rate circuit: µA741 configured as voltage follower (unity gain) with R1 = 10 KΩ, input from A.F. Oscillator (Vm = 3V). Output observed on CRO (CH2).

5. OBSERVATIONS AND CALCULATIONS

5.1 Inverting Bias Current (IB1)



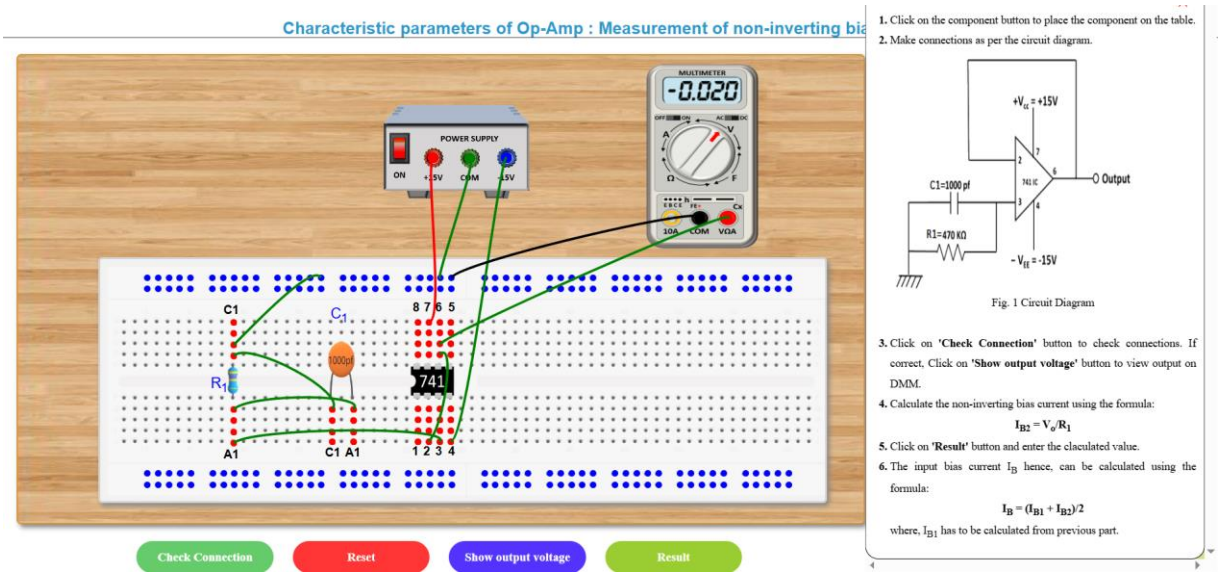
DMM Reading (Vo1): 0.024 V

Feedback Resistor Rf: 470 KΩ = 470,000 Ω

Formula: IB1 = Vo1 / Rf

IB1 = 0.024 / 470,000 = 5.1064e-8 A = 51.0638 nA

5.2 Non-Inverting Bias Current (IB2)



DMM Reading (Vo2): −0.020 V (magnitude: 0.020 V)

Resistor R1: 470 KΩ = 470,000 Ω

Formula: IB2 = |Vo2| / R1

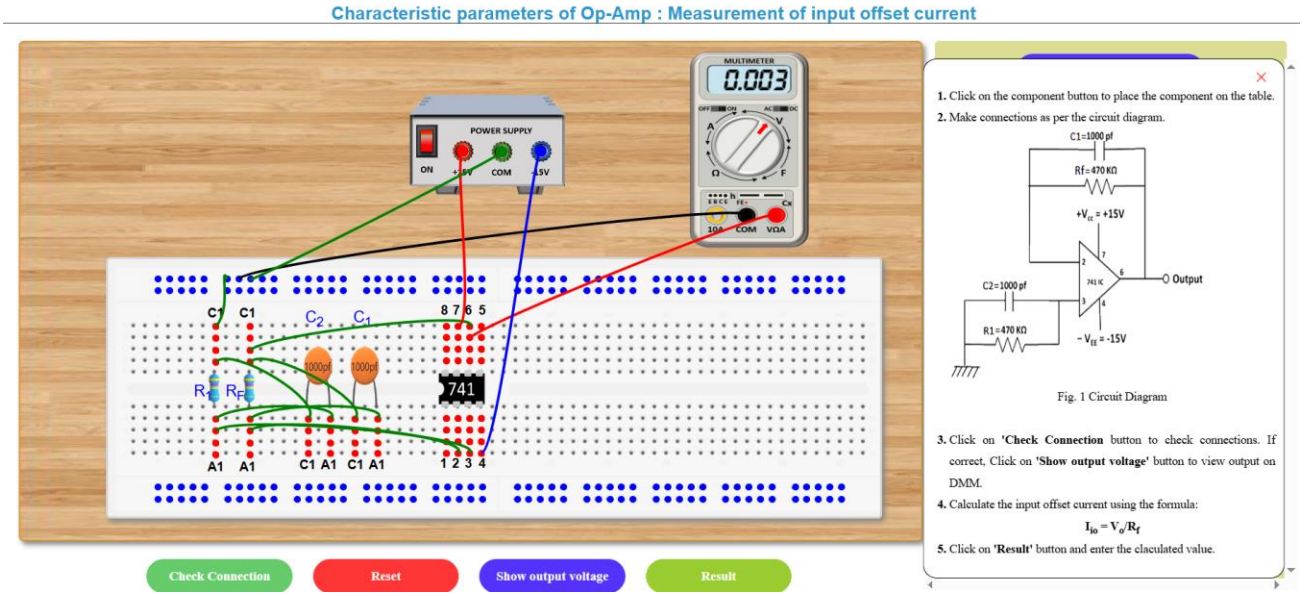
IB2 = 0.020 / 470,000 = 4.2553e-8 A = 42.5532 nA

5.3 Input Bias Current (IB)

Formula: IB = (IB1 + IB2) / 2

IB = (51.0638 + 42.5532) / 2 = 46.8085 nA ≈ 46.81 nA

5.4 Input Offset Current (IOS)

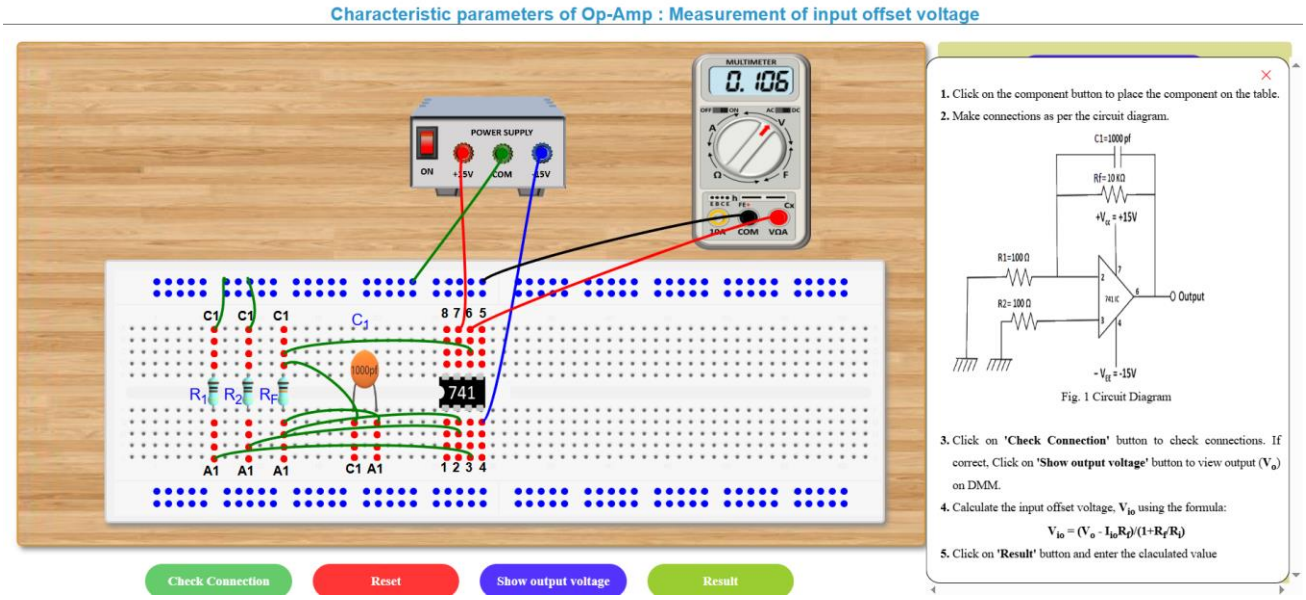


Formula: IOS = |IB1 − IB2|

IOS = |51.0638 − 42.5532| = 8.5106 nA ≈ 8.51 nA

Verification from offset current circuit: DMM = 0.003 V, Rf = 470 KΩ → Iio = 0.003/470,000 = 6.3830 nA

5.5 Input Offset Voltage (Vio)



DMM Reading (Vo4): 0.106 V

Rf: 10 KΩ = 10,000 Ω | **R1 = R2:** 100 Ω each

Step 1 – Estimate Ios contribution: Ios × Rf = (Vo4/Rf) × Rf = Vo4 = 0.106 V (from offset current circuit context)

Step 2 – Apply formula: Vio = (Vo − Ios×Rf) / (1 + Rf/R1)

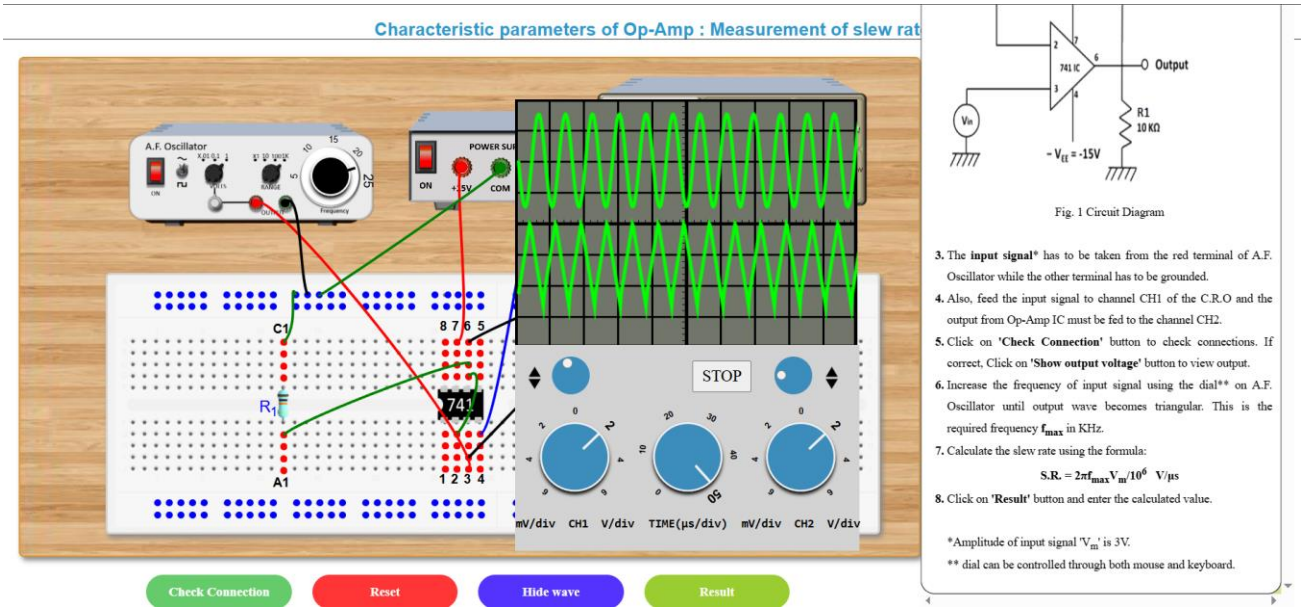
1 + Rf/R1 = 1 + (10,000/100) = 1 + 100 = 101 (gain = 101)

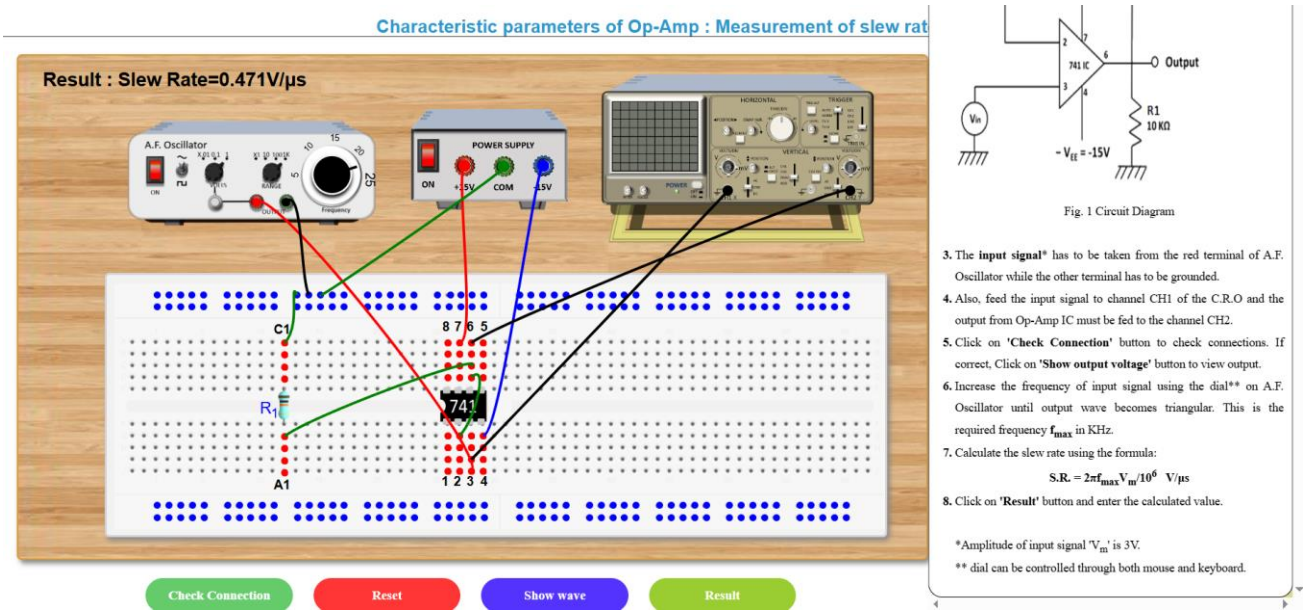
Since **Ios × Rf** is very small (nA × kΩ → μV range) compared to Vo4, it is negligible:

Vio ≈ Vo4 / (1 + Rf/R1) = 0.106 / 101 = 1.0495 mV ≈ 1.050 mV

Output Offset Voltage (VOOS): 0.106 V (directly read from DMM)

5.6 Slew Rate





Formula used: SR = 2π × fmax × Vm / 10^6 V/µs (where Vm = 3V)

calculation of SR = 2π × 25 × 3 / 10^6 V/µs

Result from virtual lab: SR = 0.471 V/µs

6. OBSERVATION TABLE

Sr. No.	Parameter	Datasheet Value (Typical/Max)	Measured Value	Status
1	**Input Bias Current (IB)**	80 nA (typ) / 1500 nA (max)	**46.81 nA**	✓Within Limits
2	**Input Offset Current (IOS)**	20 nA (typ) / 200 nA (max)	**8.51 nA**	✓Within Limits
3	**Input Offset Voltage (Vio)**	2 mV (typ) / 6 mV (max)	**1.050 mV**	✓Within Limits
4	**Output Offset Voltage (VOOS)**	Not specified directly	**0.106 V**	Observed
5	**Slew Rate (SR)**	0.5 V/µs (typ)	**0.471 V/µs**	✓Close to Datasheet

7. RESULT

The following parameters of the µA741 Op-Amp IC were successfully measured using virtual lab simulation:

Parameter	Result
Input Bias Current (IB)	46.81 nA
Input Offset Current (IOS)	8.51 nA
Input Offset Voltage (Vio)	1.050 mV
Output Offset Voltage (VOOS)	0.106 V
Slew Rate	0.471 V/µs

8. INFERENCE

Input Bias Current: The measured value of 46.81 nA is well within the datasheet typical value of 80 nA and the maximum limit of 1500 nA, confirming proper operation of the internal differential amplifier pair.

Input Offset Current: The measured IOS of 8.51 nA is well below the typical datasheet value of 20 nA and the maximum limit of 200 nA, indicating excellent transistor matching within the IC.

Input Offset Voltage: The calculated Vio of 1.050 mV is significantly below the typical value of 2 mV and the maximum of 6 mV. This confirms minimal internal imbalance in the op-amp.

Output Offset Voltage: The output offset voltage of 0.106 V observed at the output (pin 6) is consistent with the amplified input offset voltage. This value validates the relationship VOOS = Vio × (1 + Rf/R1) for the given gain of 101.

Slew Rate: The measured slew rate of 0.471 V/µs is very close to the datasheet typical value of 0.5 V/µs (a deviation of ~5.8%). This confirms the limited large-signal bandwidth of the µA741C, which restricts its use at higher frequencies.

All parameters are within the acceptable limits as per the µA741 datasheet. Experiment successfully completed via Virtual Lab.